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Administration Linux

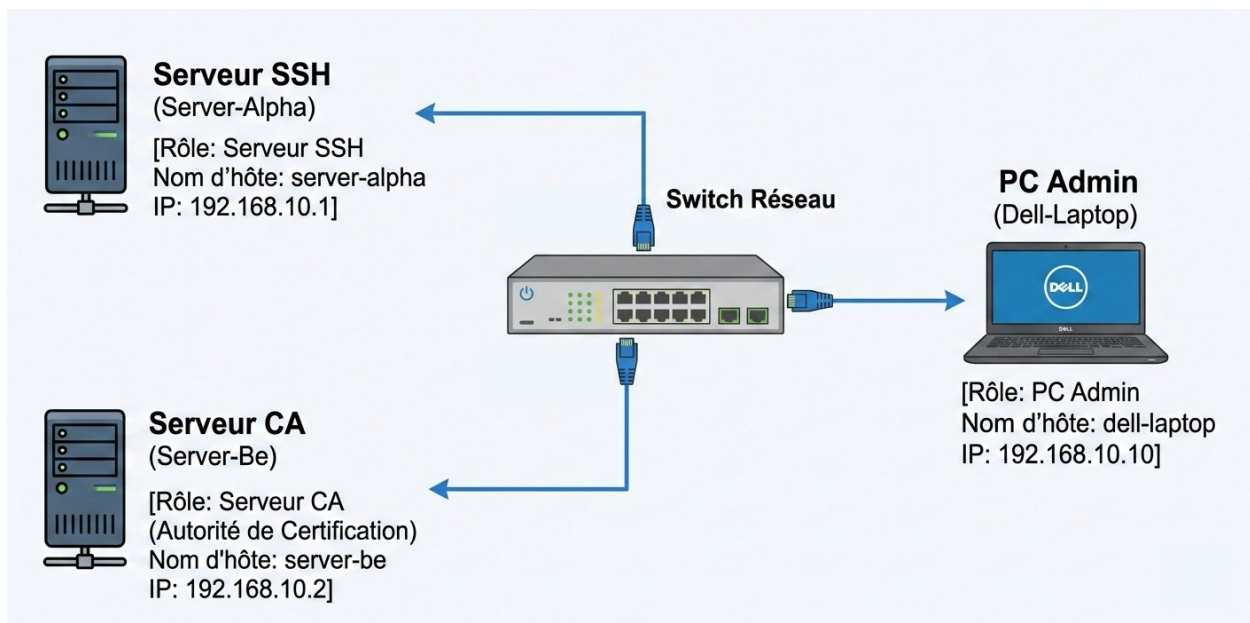
SSH SFTP Fail2ban CA

Objectif

Ce TP a pour objectif de mettre en place une authentification SSH sécurisée et centralisée. On travaille sur trois points principaux :

- Clés SSH : génération et utilisation des clés pour l'accès aux serveurs.
- CA SSH : création d'une autorité de certification pour signer les clés des utilisateurs.
- Suppression des mots de passe : connexion uniquement via clés signées pour renforcer la sécurité et faciliter l'administration.

Architecture



1. Installation du serveur SSH

On installe le paquet SSH :

```
sudo apt install openssh-server -y
```

On vérifie l'état du service :

```
sudo systemctl status ssh
```

On modifie le fichier `/etc/ssh/sshd_config` pour changer le port :

Port 2222

On redémarre le service :

```
sudo systemctl restart ssh
```

2. Installation du serveur SFTP

On active le sous-système SFTP intégré à SSH.

On édite le fichier `/etc/ssh/sshd_config` et on ajoute :

```
Subsystem sftp internal-sftp
Match User user1
    ChrootDirectory /home/user1/SFTP-backup
    ForceCommand internal-sftp
Match User user2
    ChrootDirectory /home/user2/SFTP-backup
    ForceCommand internal-sftp
```

On redémarre SSH :

```
sudo systemctl restart ssh
```

3. Création des comptes utilisateurs

On crée deux comptes utilisateurs :

```
sudo useradd -m user1
```

```
sudo useradd -m user2
```

```
hosniadmin@SERVER-ALPHA:~$ sudo useradd -m -s /bin/bash user1
hosniadmin@SERVER-ALPHA:~$ sudo useradd -m -s /bin/bash user2
hosniadmin@SERVER-ALPHA:~$ sudo passwd user1
New password:
Retype new password:
passwd: password updated successfully
hosniadmin@SERVER-ALPHA:~$ sudo passwd user2
New password:
Retype new password:
passwd: password updated successfully
hosniadmin@SERVER-ALPHA:~$ █
```

On crée les répertoires de sauvegarde :

```
sudo mkdir -p /home/user1/SFTP-backup
```

```
sudo mkdir -p /home/user2/SFTP-backup
```

On ajuste les permissions :

```
sudo chown root:root /home/user1
```

```
sudo chown root:root /home/user2
```

```
sudo chmod 755 /home/user1
```

```
sudo chmod 755 /home/user2
```

```
sudo chown user1:user1 /home/user1/SFTP-backup
```

```
sudo chown user2:user2 /home/user2/SFTP-backup
```

```
hosniadmin@SERVER-ALPHA:~$ sudo mkdir -p /home/user1/SFTP-backup
hosniadmin@SERVER-ALPHA:~$ sudo mkdir -p /home/user2/SFTP-backup
hosniadmin@SERVER-ALPHA:~$ sudo chown root:root /home/user1
hosniadmin@SERVER-ALPHA:~$ sudo chown root:root /home/user2
hosniadmin@SERVER-ALPHA:~$ sudo chmod 755 /home/user1 /home/user2
hosniadmin@SERVER-ALPHA:~$ sudo chown user1:user1 /home/user1/SFTP-backup
hosniadmin@SERVER-ALPHA:~$ sudo chown user2:user2 /home/user2/SFTP-backup
hosniadmin@SERVER-ALPHA:~$ █
```

4. Test des connexions SSH

Depuis la machine admin (IP 192.168.10.10), on teste la connexion SSH avec mot de passe :

```
ssh -p 2222 user1@192.168.10.1
```

```
ssh -p 2222 user2@192.168.10.1
```

```

hosni@dell-laptop ~$ ssh -p 2222 user2@192.168.10.1
user2@192.168.10.1's password:
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 6.8.0-87-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

Expanded Security Maintenance for Applications is not enabled.

242 updates can be applied immediately.
96 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

The programs included with the Ubuntu system are free software;
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applicable law.

user2@SERVER-ALPHA:~$

```

On vérifie que la connexion fonctionne.

On génère une paire de clés sur la machine admin :

ssh-keygen

```

* hosni@dell-laptop ~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/hosni/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/hosni/.ssh/id_ed25519
Your public key has been saved in /home/hosni/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:VMdp1E6Ngwfnu95xU9kTF99G4YHz/Oe7kB9FRlo73DE hosni@dell-laptop
The key's randomart image is:
+---[ED25519 256]---+
|      .E=|
|      oo=X|
|      . . =*O|
|      . * O **|
|      S O B .O=|
|      + = O. *|
|      . = oo.+O|
|      O   ooo+|
|      .O .+O|
+----[SHA256]-----+
hosni@dell-laptop ~$

```

On copie les clés vers le serveur :

ssh-copy-id -p 2222 user1@192.168.10.1

```

X hosni@dell-laptop ~$ ssh-copy-id -p 2222 user1@192.168.10.1

/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
user1@192.168.10.1's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh -p 2222 'user1@192.168.10.1'"
and check to make sure that only the key(s) you wanted were added.

```

ssh-copy-id -p 2222 user2@192.168.10.1

```

hosni@dell-laptop ~$ ssh-copy-id -p 2222 user2@192.168.10.1

/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
user2@192.168.10.1's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh -p 2222 'user2@192.168.10.1'"
and check to make sure that only the key(s) you wanted were added.

```

On teste la connexion SSH par clé :

ssh -p 2222 user1@192.168.10.1

ssh -p 2222 user2@192.168.10.1

```

hosni@dell-laptop ~$ ssh -p 2222 user1@192.168.10.1
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 6.8.0-87-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

Expanded Security Maintenance for Applications is not enabled.

242 updates can be applied immediately.
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Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Wed Feb 18 13:28:06 2026 from 192.168.0.105
user1@SERVER-ALPHA:~$ _

```

On désactive l'authentification par mot de passe :

sudo nano /etc/ssh/sshd_config

PasswordAuthentication no

On redémarre le service :

sudo systemctl restart ssh

5. Test SFTP

On teste l'accès SFTP pour chaque utilisateur :

```
sftp -P 2222 user1@192.168.10.1
```

```
sftp -P 2222 user2@192.168.10.1
```

```
hosni@dell-laptop ~$ sftp -P 2222 user1@192.168.10.1
The authenticity of host '[192.168.10.1]:2222 ([192.168.10.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:4xEcl+jElm+92oMwofzYdm6EIO+m448E4w6KGHFluko.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[192.168.10.1]:2222' (ED25519) to the list of known hosts.
user1@192.168.10.1's password:
Connected to 192.168.10.1.
sftp> help
Available commands:
bye                Quit sftp
cd path            Change remote directory to 'path'
chgrp [-h] grp path Change group of file 'path' to 'grp'
chmod [-h] mode path Change permissions of file 'path' to 'mode'
chown [-h] own path Change owner of file 'path' to 'own'
copy oldpath newpath Copy remote file
cp oldpath newpath  Copy remote file
```

```
hosni@dell-laptop ~$ sftp -P 2222 user2@192.168.10.1
Connected to 192.168.10.1.
sftp> _
```

On vérifie que chaque utilisateur est bien limité à son répertoire SFTP-backup.

6. Installation et configuration de Fail2Ban

On installe le paquet :

```
sudo apt install fail2ban -y
```

```
hosniadmin@SERVER-ALPHA:~/Desktop$ sudo apt install fail2ban -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libwpe-1.0-1 libwpebackend-fdo-1.0-1
Use 'sudo apt autoremove' to remove them.
Suggested packages:
  mailx monit sqlite3
The following NEW packages will be installed:
  fail2ban
0 upgraded, 1 newly installed, 0 to remove and 235 not upgraded.
Need to get 394 kB of archives.
After this operation, 2,098 kB of additional disk space will be used.
0% [Connecting to ubuntu-mirror-2.ps6.canonical.com]
```

On copie le fichier de configuration par défaut :

```
sudo cp /etc/fail2ban/jail.conf /etc/fail2ban/jail.local
```

On édite /etc/fail2ban/jail.local et on active la protection SSH :

```
[sshd]
enabled = true
port = 2222
filter = sshd
logpath = /var/log/auth.log
maxretry = 3
bantime = 1h
```

```
#
[sshd]
# To use more aggressive sshd modes set filter parameter "mode" in jail.local:
# normal (default), ddos, extra or aggressive (combines all).
# See "tests/files/logs/sshd" or "filter.d/sshd.conf" for usage example and details.
#mode = normal
enabled
port = 2222
filter = sshd
logpath = %(sshd_log)s
backend = %(sshd_backend)s
maxretry = 3
bantime = 1h
```

On redémarre le service :

```
sudo systemctl restart fail2ban
```

On vérifie le statut :

```
fail2ban-client status sshd
```

```
hosniadmin@SERVER-ALPHA:~/Desktop$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 2
| |- Total failed: 4
| \- File list: /var/log/auth.log
- Actions
  |- Currently banned: 0
  |- Total banned: 0
  \- Banned IP list:
hosniadmin@SERVER-ALPHA:~/Desktop$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 0
| |- Total failed: 5
| \- File list: /var/log/auth.log
- Actions
  |- Currently banned: 1
  |- Total banned: 1
  \- Banned IP list: 192.168.0.102
hosniadmin@SERVER-ALPHA:~/Desktop$
```

7. Configuration du serveur CA

7.1. Mise en place de l'autorité de certification SSH

Sur le serveur ca (**192.168.10.2**), on crée le répertoire qui va contenir la clé de l'autorité de certification.

```
sudo mkdir -p /etc/ssh/ca
```

```
cd /etc/ssh/ca
```

On génère la clé du CA :

```
sudo ssh-keygen -t rsa -f /etc/ssh/ca/ssh_ca
```

```
hosniadmin@SERVER-BE:/etc/ssh/ca$ sudo ssh-keygen -t rsa -f /etc/ssh/ca/ssh_ca
Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /etc/ssh/ca/ssh_ca
Your public key has been saved in /etc/ssh/ca/ssh_ca.pub
The key fingerprint is:
SHA256:BRDfq0be8yoJZeu51o2GIcECnlqCiq3MxEVR55If+a8 root@SERVER-BE
The key's randomart image is:
+---[RSA 3072]-----+
|  o+o+o.          |
|. 000.. +         |
|o.o...o o o       |
|000.o.00 .        |
|=+.. +o.S         |
|+o. 000 .         |
|+.  +.*.o         |
|.o  .BE= .        |
|  .o+o+.         |
+---[SHA256]-----+
```

Deux fichiers sont créés :

- ssh_ca → clé privée du CA (doit rester sur le serveur CA)
- ssh_ca.pub → clé publique du CA

```
hosniadmin@SERVER-BE:/etc/ssh/ca$ ls -l
total 8
-rw----- 1 root root 2655 Mar  8 16:14 ssh_ca
-rw-r--r-- 1 root root  568 Mar  8 16:14 ssh_ca.pub
```

7.2. Distribution de la clé publique du CA

On copie la clé publique vers le serveur SSH.

Depuis le serveur ca:

```
scp -P 2222 /etc/ssh/ca/ssh_ca.pub user@192.168.10.1:/tmp/
```

```
hosniadmin@SERVER-BE:/etc/ssh/ca$ scp -P 2222 /etc/ssh/ca/ssh_ca.pub hosniadmin@192.168.10.1:/tmp/
The authenticity of host '[192.168.10.1]:2222 ([192.168.10.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:4xEcl+jElm+92oMwofzYdm6EIO+m448E4w6KGHFluko.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[192.168.10.1]:2222' (ED25519) to the list of known hosts.
hosniadmin@192.168.10.1's password:
Permission denied, please try again.
hosniadmin@192.168.10.1's password:
ssh_ca.pub                               100% 568    322.1KB/s   00:00
```

Sur le serveur ssh (**192.168.10.1**) :

```
sudo mkdir -p /etc/ssh/ca
```

```
sudo mv /tmp/ssh_ca.pub /etc/ssh/ca/ssh_ca.pub
```

```
sudo chmod 644 /etc/ssh/ca/ssh_ca.pub
```

```
hosniadmin@SERVER-ALPHA:~/Desktop$ sudo mkdir /etc/ssh/ca
hosniadmin@SERVER-ALPHA:~/Desktop$ sudo mv /tmp/ssh_ca.pub /etc/ssh/ca/
hosniadmin@SERVER-ALPHA:~/Desktop$ sudo chmod 644 /etc/ssh/ca/ssh_ca.pub
hosniadmin@SERVER-ALPHA:~/Desktop$
```

On modifie la configuration SSH :

```
sudo nano /etc/ssh/sshd_config
```

On ajoute la ligne :

```
TrustedUserCAKeys /etc/ssh/ca/ssh_ca.pub
```

On vérifie la configuration :

```
sudo sshd -t
```

On redémarre le service SSH :

```
sudo systemctl restart ssh
```

Le serveur SSH accepte maintenant les certificats signés par ce CA.

7.3. Génération des clés utilisateurs

Sur le pc admin (**192.168.10.10**) on génère les clés pour les utilisateurs.

Pour user1 :

```
ssh-keygen -t rsa -f ~/.ssh/id_user1
```

```
hosni@dell-laptop ~$ ssh-keygen -t rsa -f ~/.ssh/lab/id_user1

Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/hosni/.ssh/lab/id_user1
Your public key has been saved in /home/hosni/.ssh/lab/id_user1.pub
The key fingerprint is:
SHA256:MIE3lia02uTmh8DrTNL6bGmSYaU8lGYQzetwtgnr27Q hosni@dell-laptop
The key's randomart image is:
+---[RSA 3072]-----+
|. + ..... |
|. o o.*. |
|. ..O=O. |
|O**+  o |
|=B*o+  S |
|O**=  . |
|+O=OO  . |
|oO*  .. |
|. **E   |
+-----[SHA256]-----+
```

Pour user2 :

```
ssh-keygen -t rsa -f ~/.ssh/id_user2
```

```
hosni@dell-laptop ~$ ssh-keygen -t rsa -f ~/.ssh/lab/id_user2

Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/hosni/.ssh/lab/id_user2
Your public key has been saved in /home/hosni/.ssh/lab/id_user2.pub
The key fingerprint is:
SHA256:E/bL36nhE1n/CfpkpICs/8ZV9RN4MILfkIdn1hU8Ssc hosni@dell-laptop
The key's randomart image is:
+---[RSA 3072]-----+
|      ..OO*.*+ |
|      . +.B.E |
|      o . O = + |
|      ...O . +... |
|      oS.. .+ .. |
|      . O.O=. . |
|      . . +.O+. O |
|      . O ++O O. |
|      .O. =+O |
+-----[SHA256]-----+
```

Les fichiers créés sont :

- id_user1
- id_user1.pub
- id_user2
- id_user2.pub

```
hosni@dell-laptop ~$ ls ~/.ssh/lab/
id_user1  id_user1.pub  id_user2  id_user2.pub
```

7.4. Envoi des clés publiques au CA

On copie les clés publiques des utilisateurs depuis le pc admin vers le serveur ca :

```
hosniadmin@SERVER-BE:~$ ls /tmp
id_user1.pub
id_user2.pub
snap-private-tmp
systemd-private-4a87f78e8ad54576a443eef36e3edac3-colord.service-EI6GI3
```

7.5. Signature des clés utilisateurs

Sur le serveur ca, on signe les clés publiques avec la clé du CA :

```
ssh-keygen -s /etc/ssh/ca/ssh_ca -I user1 -n user1
/tmp/id_user1.pub
```

```
hosniadmin@SERVER-BE:~$ sudo ssh-keygen -s /etc/ssh/ca/ssh_ca -I user1 -n user1 /tmp/id_user1.pub
Enter passphrase:
Signed user key /tmp/id_user1-cert.pub: id "user1" serial 0 for user1 valid forever
```

```
ssh-keygen -s /etc/ssh/ca/ssh_ca -I user2 -n user2
/tmp/id_user2.pub
```

```
hosniadmin@SERVER-BE:~$ sudo ssh-keygen -s /etc/ssh/ca/ssh_ca -I user2 -n user2 /tmp/id_user2.pub
Enter passphrase:
Signed user key /tmp/id_user2-cert.pub: id "user2" serial 0 for user2 valid forever
```

Les certificats générés sont :

- id_user1-cert.pub
- id_user2-cert.pub

```

hosniadmin@SERVER-BE:~$ ls /tmp/
id_user1-cert.pub
id_user1.pub
id_user2-cert.pub
id_user2.pub
ssap-private.tmp

```

7.6.Retour des certificats vers le pc admin

On copie les certificats signés vers le pc admin.

Les certificats sont maintenant dans le dossier .ssh du client.

```

hosni@dell-laptop ~$ ls ~/.ssh/lab/
id_user1  id_user1-cert.pub  id_user1.pub  id_user2  id_user2-cert.pub  id_user2.pub

```

7.7.Tests de connexion avec certificat

Depuis le pc admin, on teste la connexion.

Pour user1 :

```
ssh -p 2222 -i ~/.ssh/id_user1 user1@192.168.10.1
```

```

(user1) 192.168.10.1:2222
hosni@dell-laptop ~$ ssh -p 2222 -i ~/.ssh/lab/id_user1 user1@192.168.10.1
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 6.8.0-87-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

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Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Sun Mar  8 16:52:23 2026 from 192.168.10.100
user1@SERVER-ALPHA:~$

```

Pour user2 :

```
ssh -p 2222 -i ~/.ssh/id_user2 user2@192.168.10.1
```

```
(user2) 192.168.10.1:2222

Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 6.8.0-87-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

Expanded Security Maintenance for Applications is not enabled.

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The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

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individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

Last login: Sun Mar  8 16:53:12 2026 from 192.168.10.100
user2@SERVER-ALPHA:~$ _
```

La connexion doit s'établir sans mot de passe grâce au certificat signé par le CA.

7.8.Vérification sans certificat

On renomme temporairement le certificat :

```
mv ~/.ssh/id_user1-cert.pub ~/.ssh/id_user1-cert.pub.bak
```

On tente une connexion :

```
ssh -p 2222 -i ~/.ssh/id_user1 user1@192.168.10.1
```

La connexion doit échouer.

On restaure ensuite le certificat :

```
mv ~/.ssh/id_user1-cert.pub.bak ~/.ssh/id_user1-cert.pub
```

7.9.Test SFTP

On teste également l'accès SFTP :

```
sftp -P 2222 user1@192.168.10.1
```

```
Connection to 192.168.10.1 closed.  
hosni@dell-laptop ~$ sftp -P 2222 user1@192.168.10.1  
Connected to 192.168.10.1.  
sftp> _
```

```
sftp -P 2222 user2@192.168.10.1
```

L'accès doit fonctionner avec le certificat.